

ORACLE PLSQL

PL/SQL Architecture and Basics

- *Definition:* PL/SQL stands for Procedural Language extension to SQL. It is an Oracle proprietary language that allows you to add procedural logic—such as conditions (IF) and loops—to standard SQL statements.
- *Architecture:* When a PL/SQL block is submitted, the PL/SQL Engine within the database server segregates the code. Procedural statements are sent to the Procedural Statement Executor, while SQL statements are sent to the SQL Statement Executor.
- *Context Switching:* The interaction between these two executors is called context switching. Frequent switching (e.g., in a large loop) can cause performance issues.
- *Block Structure:* PL/SQL uses a block structure consisting of:
 - *DECLARE:* Optional section for variable and constant definitions.
 - *BEGIN...END:* Mandatory section for executable statements.
 - *EXCEPTION:* Optional section for error handling.

Variables, Constants, and Data Types

- *Assignments:* Use the assignment operator (:=) to give values to variables.
- *Constants:* Using the *CONSTANT* keyword prevents a variable's value from being changed after initialization.
- *Dynamic Data Types:*
 - *%TYPE:* Used to declare a variable with the same data type as a specific table column.
 - *%ROWTYPE:* Used to declare a variable (a record) that represents an entire row of a table.

Control Statements and Loops

- *Conditions:* The video covers *IF-THEN-ELSEIF-ELSE* structures and *CASE* statements for branching logic.
- *Loops:*
 - *Simple Loop:* Executes until an *EXIT WHEN* condition is met.
 - *WHILE Loop:* Checks a condition before entering the loop.
 - *FOR Loop:* Iterates through a specified range of integers.

Cursors

Cursors are private work areas used to execute SQL and handle multiple rows.

- *Implicit Cursors:* Automatically managed by Oracle for DML statements and `SELECT INTO`.
- *Explicit Cursors:* User-defined for multi-row queries following four steps: Declare, Open, Fetch, and Close.
- *Attributes:* Used to check the state of a cursor, such as `%FOUND`, `%NOTFOUND`, `%ROWCOUNT`, and `%ISOPEN`.
- *Ref Cursors:* Variables that point to a query result, which can be Strongly Typed (with a return type) or Weakly Typed.

Named Blocks (Stored Objects)

Unlike anonymous blocks, named blocks are stored as objects in the database.

- *Procedures:* Used for logical data flow and computations. They support `IN`, `OUT`, and `IN OUT` parameters.
- *Functions:* Similar to procedures but must return a value using the `RETURN` keyword. They can be called directly within SQL statements.
- *Packages:* Containers that group related procedures and functions. They consist of a Specification (publicly visible declarations) and a Body (private implementation).
- *Triggers:* Stored programs that automatically fire in response to events like `INSERT`, `UPDATE`, or `DELETE`. Types include DML, DDL, and "Instead Of" triggers for complex views.

Performance Tuning and Advanced Concepts

- *Bulk Collect & FORALL:* These features reduce context switching by processing data in bulk rather than row-by-row, significantly improving performance.
- *Dynamic SQL:* Using `EXECUTE IMMEDIATE` allows you to execute SQL strings built at runtime (e.g., dropping a table dynamically).
- *Exception Handling:* Used to manage runtime errors. Oracle provides Predefined Exceptions (like `NO_DATA_FOUND`), or users can define their own.

- *Autonomous Transactions: Using `PRAGMA AUTONOMOUS_TRANSACTION` allows a subprogram to commit or roll back its own changes independently of the main transaction.*
- *Temporary Tables: The video discusses Global Temporary Tables (data persistent for the session or until commit) and Private Temporary Tables.*